$$P_4 + 3O_2 \xrightarrow{\Delta} \underset{\text{Phosphoroustrioxide}}{P_4O_6}$$

$$P_4 + 5O_2 \xrightarrow{\Delta} \underset{\text{Phosphorouspentoxide}}{P_4O_{10}}$$

Reaction with chlorine: Phosphorus reacts with chlorine to form tri and penta chloride. Yellow phosphorus reacts violently at room temperature, while red phosphorus reacts on heating

$$P_4 + 6Cl_2 \longrightarrow \underset{\text{Phosphorous tri chloride}}{4PCl_3}$$

$$P_4 + 10Cl_2 \longrightarrow \underset{\text{Phosphorous penta chloride}}{4PCl_5}$$

Reaction with alkali: Yellow phosphorus reacts with alkali on boiling in an inert atmosphere liberating phosphine. Here phosphorus act as reducing agent.

$$P_4 + 3NaOH + 3H_2O \longrightarrow \underset{\text{sodium hypo phosphite}}{3NaH_2PO_2} + \underset{\text{Phosphine}}{PH_3}\uparrow$$

Reaction with nitric acid: When phosphorus is treated with conc. nitric acid it is oxidised to phosphoric acid. This reaction is catalysed by iodine crystals.

$$P_4 + 20HNO_3 \longrightarrow \underset{\text{Ortho phosphoric acid}}{4H_3PO_4} + 20NO_2 + 4H_2O$$

Reaction with metals: Phosphorus reacts with metals like Ca and Mg to give phosphides.. Metals like sodium and potassium react with phosphorus vigorously.

$$P_4 + 6Mg \longrightarrow \underset{\text{Magnesium phosphide}}{2Mg_3P_2}$$

$$P_4 + 6Ca \longrightarrow \underset{\text{Calcium phosphide}}{2Ca_3P_2}$$

$$P_4 + 12Na \longrightarrow \underset{\text{Sodium phosphide}}{4Na_3P} \ .$$

Uses of phosphorus:

1. The red phosphorus is used in the match boxes
2. It is also used for the production of certain alloys such as phosphor bronze

3.1.9 Phosphine (PH_3)

Phosphine is the most important hydride of phosphorus

Preparation:

Phosphine is prepared by action of sodium hydroxide with white phosphorus in an inert atmosphere of carbon dioxide or hydrogen.

$$P_4 + 3NaOH + 3H_2O \longrightarrow \underset{\text{sodium hypo phosphite}}{3NaH_2PO_2} + \underset{\text{Phosphine}}{PH_3}\uparrow$$

Phosphine is freed from phosphine dihydride(P_2H_4) by passing through a freezing mixture. The dihydride condenses while phosphine does not.

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